VAPO: Value-Model-Based RL for Advanced Reasoning

60.4 on AIME 2024

ByteDance Seed | Base Model: Qwen2.5-32B

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought

Value-Free (GRPO/DAPO)

Computes advantage based on final trajectory reward and assigns it **uniformly across all tokens**.

Value-Based (VAPO)

Enables **precise token-level credit assignment**, tracing each action's impact on returns.

*Critical when subtle step-level errors cause catastrophic failures in multi-step reasoning.

Three Core Challenges

Plaguing Value-Model-Based RL in Long-CoT Tasks

01 / Value Model Bias

Arises from excessively long trajectories and the compounded errors of bootstrapped learning.

02 / Heterogeneous Lengths

Varying reasoning lengths require completely different bias-variance tradeoffs during GAE.

03 / Reward Signal Sparsity

In long reasoning chains, only the final `<eos>` token receives an actual environmental reward.

The VAPO Framework

Integrated Solutions mapping directly to the challenges.

Challenge: Value Model Bias	→	**Value-Pretraining** (Reduces initial bias)
Challenge: Heterogeneous Lengths	→	**Length-Adaptive GAE** (Dynamic λ adjustment)
Challenge: Robust Estimation	→	**Decoupled-GAE**
Additional Enhancements	→	Clip-Higher, Token-level Loss, Positive Example LM Loss, Group-Sampling

Length-Adaptive GAE

Dynamic λ Adjustment

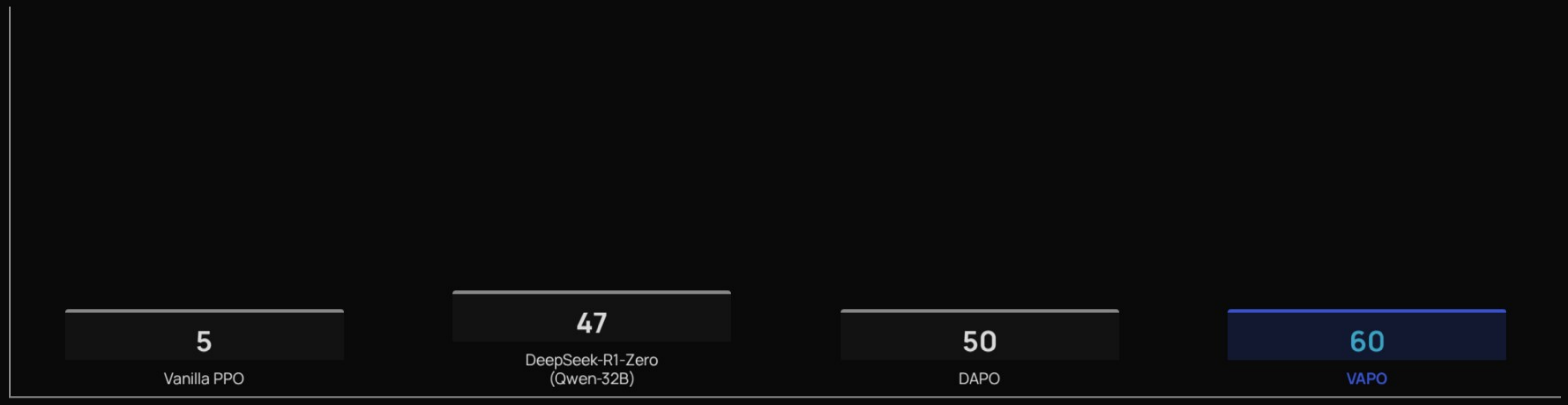
- **Short Responses:** Uses lower λ → *Lower Variance*
- **Long Responses:** Uses higher λ → *Lower Bias*

This novel mechanism gracefully handles the heterogeneous sequence lengths inherent to advanced reasoning tasks.

Sequence Length	λ Parameter	Optimizes For
Short (e.g. 500)	Low (~0.95)	Variance
Long (e.g. 6000)	High (~0.99)	Bias

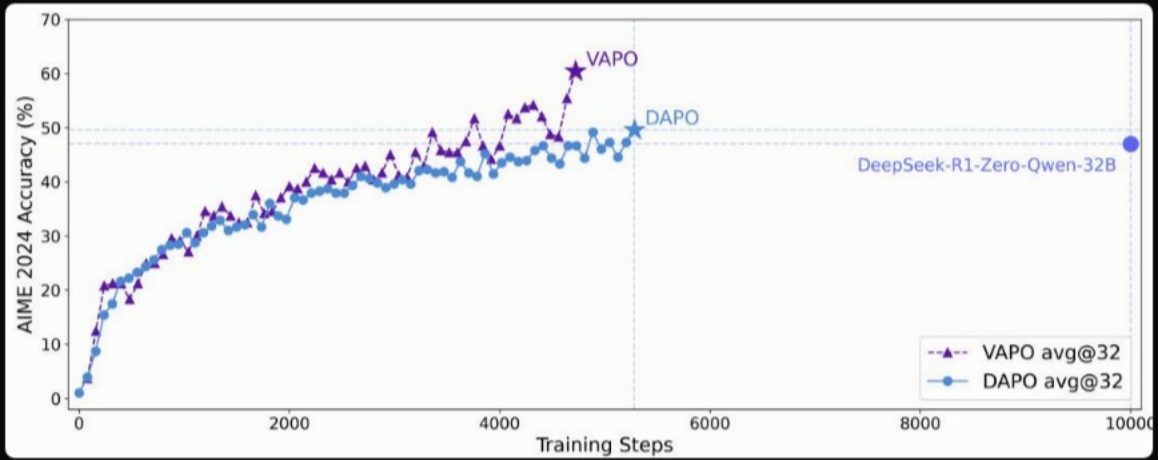
SOTA on AIME 2024

VAPO scores 60 — 10+ points above DAPO and DeepSeek-R1-Zero.

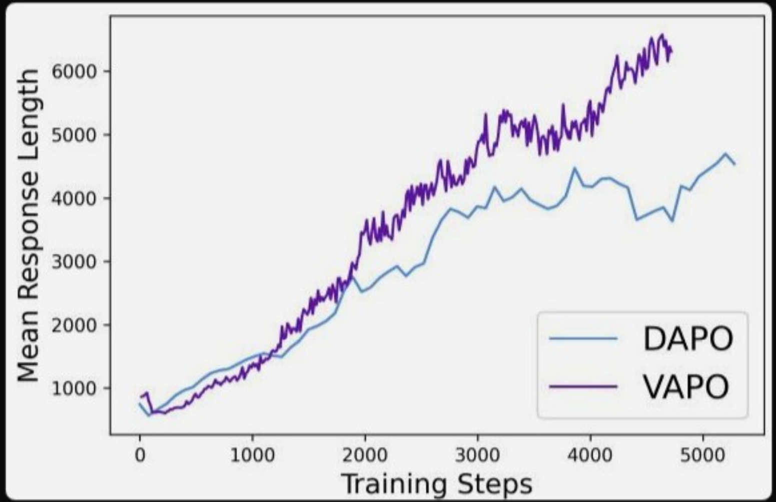


Highly Efficient & Stable

Reaches SOTA in 5K Steps — Zero Training Crashes.



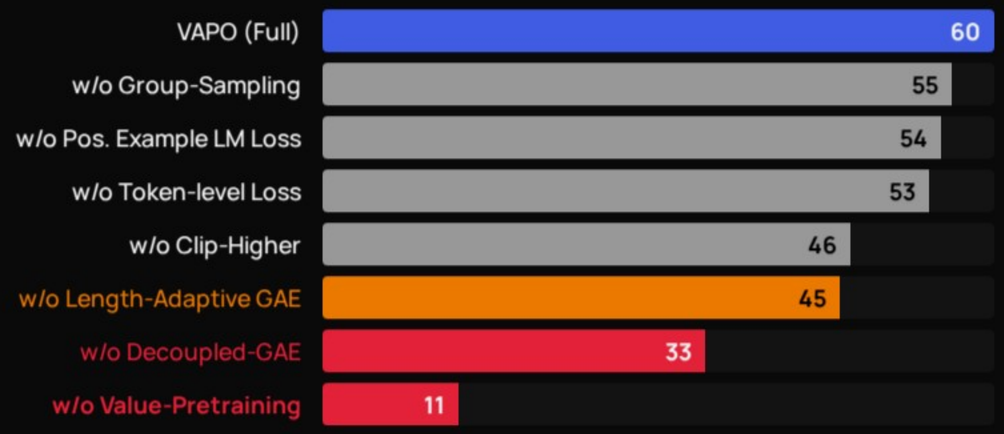
VAPO consistently outperforms DAPO throughout training.



VAPO learns longer, more elaborate chains (~6,000 tokens).

Ablation: Every Component Matters

Removing Value-Pretraining crashes the model from 60 to 11.



Key Takeaways

Value-Model-Based RL Is Back — With the Right Engineering

- **Outperforms Value-Free:** Surpasses GRPO/DAPO when bias, length, and sparsity are addressed systematically.
- **Value Pretraining is Mandatory:** The single most critical component (+49 points).
- **Length-Adaptive GAE works:** A highly effective, novel solution for variable-length generation.
- **Rock-Solid Stability:** Zero training crashes and reaches state-of-the-art in just ~5,000 steps.